

VISION — Config-First Automation

WHO

Senior Data Scientists

Freed from 60-70% boilerplate labour — refocused on model strategy, stakeholder insight, and high-value feature engineering.

TRIGGER

Scaling

Teams cannot increase throughput beyond 6-8 models/year without headcount. Framework unlocks 10-14 models/year per team.

EXPECTATION

Config-First Delivery

1-2 week cycles. 100% SHAP coverage. Auto-generated SR 11-7 model cards in <4 hours. Zero new Python per new problem type.

Executive So-What: framework adoption is the way to scale output without headcount growth.

Every Project Restarts From Zero

KPI / OBI	Measured Baseline
Project cycle time	4–8 weeks (avg 6.5 wks)
% time on boilerplate	60–70% of project hours
Algorithms tested / project	1–2 (familiarity bias)
SHAP explainability coverage	~0% standardised
SR 11-7 model card time	3–5 business days / model
Pipeline divergence (60d)	>60% of deployments
Experiment tracking	Local Excel / memory only

Executive So-What: capacity cost.

One YAML File Replaces 6.5 Weeks of Code

CURRENT STATE — Code-First Chaos

- Blank notebook per project — weeks to ramp
- 1–2 algorithms tested (familiarity bias)
- Manual HPO: no budget, no logging
- SHAP optional, ~0% coverage
- Model card: 3–5 days, inconsistent
- Training / inference pipeline diverges >60d
- Experiment results in local Excel or memory

DESIRED FUTURE STATE — Config-First Automation

- Zero new Python — YAML config only (C4)
- ≥ 8 algorithms benchmarked; champion via McNemar
- Optuna TPE: ≤ 50 trials, within 2% of manual AUC
- SHAP 100% coverage across ALL model types
- Auto-generated SR 11-7 card: <4 hours review
- Serialized pipeline: training = inference (ONNX)
- MLflow: all trials logged, staged, version-controlled

Executive So-What: The DFS converts a 6.5-week manual process into a 1-2 week config-driven pipeline — every metric is measurable and statistically testable against null hypotheses.

Four MECE Pillars

PILLAR	L0 — STRATEGIC GAP	L1 — THEMATIC GAP	L2 — ROOT CAUSE	L3 — OPERATIONAL ACTION
PROCESS	Zero reusable pipeline — 60-70% capacity on non-differentiating boilerplate	Pipeline standardisation & zero-code onboarding	No central framework ownership; preprocessing logic trapped in notebook cells	Modular Sklearn Pipeline + YAML config schema
TECHNOLOGY	No systematic model selection, HPO, or experiment tracking	Algorithm coverage & reproducible experiment tracking	Practitioner familiarity bias, no MLflow = not SR 11-7 auditable	Optuna TPE + algorithm registry + MLflow + McNemar champion selection
EXPLAINABILITY	Black-box outputs create stakeholder adoption resistance and ECOA/FCRA compliance risk	SHAP coverage 0%→100%; fairness metrics absent	SHAP KernelExplainer cost avoided; no AIF360; no ECOA adverse action mapping	Mandatory SHAP ; LIME narratives; AIF360 fairness; ECOA adverse factors
GOVERNANCE	Manual, inconsistent model cards delay production — no model inventory or version control	SR 11-7 model cards & MLflow model registry	SR 11-7 card created retrospectively; no model registry; sensitivity ratings not captured	Auto-generated model card from MLflow metadata; SR 11-7 checklist validation

L0 Strategic — Board/Exec level intent

L1 Thematic — Director level themes

L2 Analytical — Root causes

L3 Operational — Executable actions

Executive So-What: All four pillars share a common failure mode — absence of shared infrastructure. A single modular framework simultaneously closes all 16 gap cells across Process, Technology, Explainability, and Governance.